MILITARY SPECIFICATION SHEET

COMPUTER SET, STANDARD AIRBORNE
CP-1501A/AYK-14(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

The complete requirements for acquiring the computer described herein shall consist of this specification and the latest issue of MIL-C-85518.

REQUIREMENTS:

System configuration:
SRA location - See Table I.

TABLE I. SRA location in the CP-1501A computer.

SRA	Chassis Location	MIL-C-85518 Reference
-	A01	---
DIM	A02	3.6.2.2.1.3
-	A03	---
NTDS ANEW (C1)	A04	3.6.2.2.1.1.1
-	A05	---
NTDS ANEW (C1)	A06	3.6.2.2.1.1.1
IOP	A07	3.6.2.1.2
PSM	A08	3.6.2.1.1
GPM	A09	3.6.2.1.1
EAU	A10	3.6.2.1.4
MCM	A11	3.6.2.3.3
-	A12	---
-	A13	---
CMM-32	A14	3.6.2.3.1.1
CMM-32	A15	3.6.2.3.1.1
PCM-2	A16	3.6.2.4.2
Type 1 Chassis	-	3.6.2.5.1.1
Cooling Adapter	-	3.6.2.5.3

Total weight: 49.0 lbs. maximum, excluding external cables.

Dimensions: Length 19.56 inches, width 10.125 inches, height 7.625 inches excluding front panel connectors, and handles. See Figure 47 of MIL-C-85518.

Operating power: 115 volts ac, 400 Hz (3 phase (WYE), 460 Watts/540VA maximum.

Reliability in mean-time-between-failure (MTBF): MTBF shall be 870 hours minimum when tested in accordance with 4.4.3.4 of MIL-C-85518.

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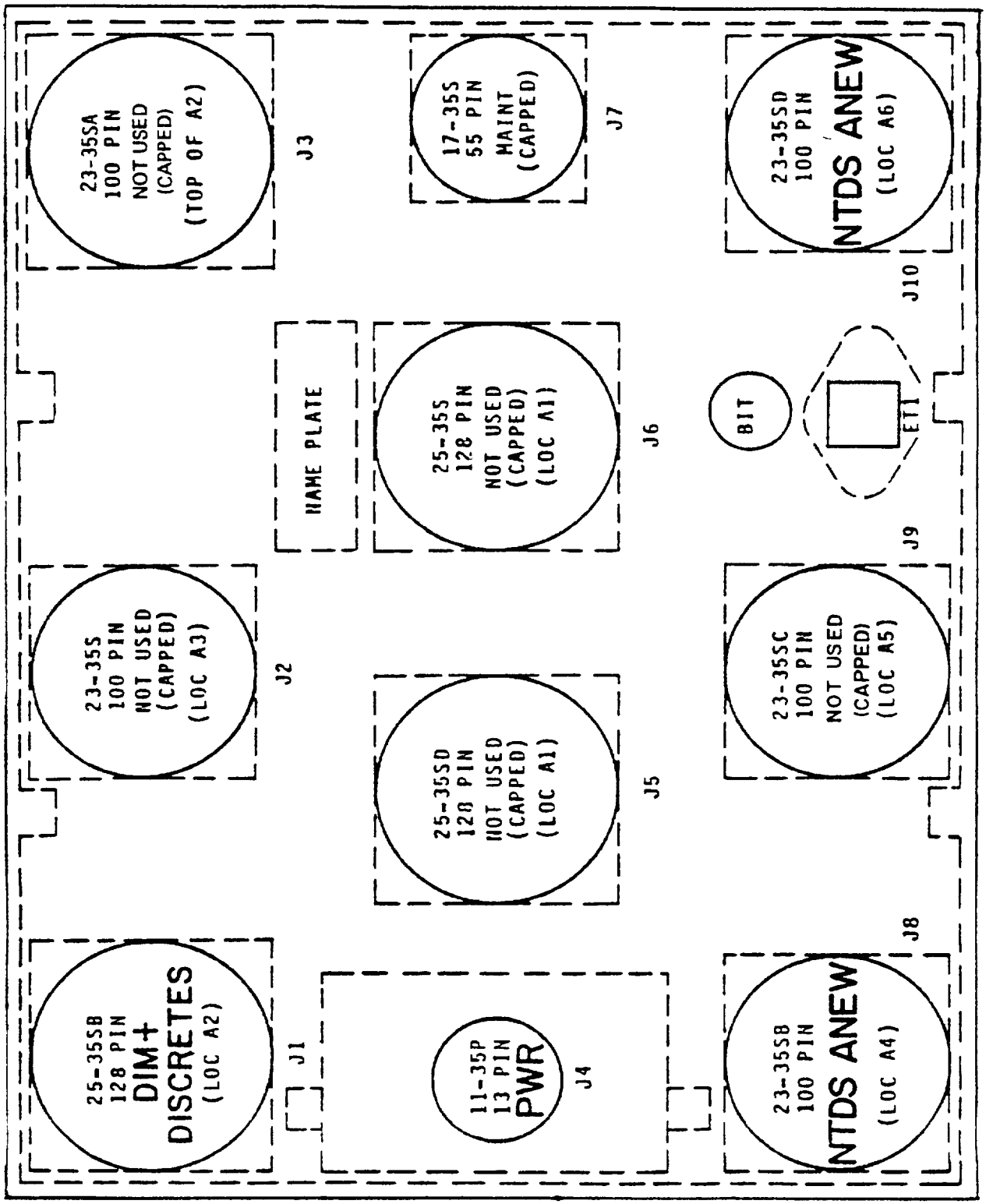


FIGURE 1. CP-1501 front panel.

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Mean-time-to-repair (MTTR): Intermediate level SRA replacement: MTTR shall be 22 minutes maximum.

Connector assignments: See Figure 1. Unused connectors shall be capped.

Service conditions: As specified in MIL-E-5400 for Class 2X equipment using the conditioned cooling air.

Heat dissipation: Electrical heat generated within the unit (Qe) shall be 460 watts maximum. Heat load due to ambient conditions (Qa) shall be 30 watts maximum.

Cooling air flow criteria:

Cooling flow: Air flow for adequate cooling based on the total specified heat dissipation (Qe + Qa) and a maximum exit cooling air temperature of 160°F (71°C) shall be not greater than 1.18 pounds per minute at an inlet cooling air temperature of 62°F (16.7°C), or 0.91 pounds per minute at 32°F (0°C).

Pressure drop: The cooling air total pressure drop for the equipment is defined as the difference between the total pressure (static plus dynamic) entering the unit and the ambient pressure surrounding the unit. The total pressure drop of this unit shall be 0.59 (±10 percent) inches of water at the rate of 1.18 pounds per minute, an inlet cooling air temperature of 62°F (16.7°C), and an ambient pressure of 14.7 psia.

Quality assurance provisions: With the exception of AET, when cooling air is required during tests specified in 4.2, 4.3, and 4.4 of MIL-C-85518, the air shall be supplied at a temperature in the range of -65°F to 120°F. The flow rate shall be adjusted to correspond to the selected inlet temperature, an exhaust temperature of 160°F, and the total heat load on the unit.

Preparing Activity
Navy AS

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